

Allocation of Decision Rights and the Investment Strategy of Mutual Funds

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Keywords: Teams, Coordination Costs, Decision Rights, Performance, Mutual Funds

JEL Codes: *G20, G23, D70*

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Abstract

The literature suggests that while decentralized decision-making can allow for greater specialization in an organization, it heightens the cost of coordinating decisions. The mutual fund industry – in particular, sole- and team-managed balanced funds – provides an ideal setting to test the specialization vs. coordination trade-off, since information on decision structures and fund actions is easily obtained. We document that sole-managed balanced funds, with centralized decision rights, exhibit significant market timing that requires reallocation across asset classes. However, consistent with coordination difficulties between managers specializing in particular asset classes, there is no market-timing evident in team-managed balanced funds. Team-managed funds exhibit greater returns from specialization, in the form of better security-selection performance than sole-managed funds. These results hold cross-sectionally and for funds that switch management structures. The overall returns across different management structures are similar, indicating a market equilibrium. Investor flows reward market-timing performance for sole- but not team-managed funds.

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1 Introduction

How should decision rights be located within organizations? What are the consequences of centralized versus decentralized decision-making? In their analysis of organizational structure, Jensen and Meckling (1992) emphasize the cost and distortion associated with communicating information within organizations. They posit that decentralized structures benefit from lower information transmission costs by giving decision rights to those with ‘specific’ information. Decentralized organizations, however, face a challenge in coordinating actions between agents that may differ in terms of their information and incentives (also see Hart and Moore, 2005; Hart and Holmstrom, 2010). These coordination problems can be alleviated in a centralized structure in which the decision-maker has a greater span of control, while having less specific information. Similar insights emerge from the literature on teams. It is suggested that while teams can add value when members have specialized and complementary skills (Lazear, 1998), there are substantial costs associated with monitoring and coordinating actions within teams (Alchian and Demsetz, 1972; Becker and Murphy, 1992).¹

Our objective in this paper is to propose and test predictions of the decision-rights theory. We test whether centralized decision structures implement strategies consistent with: (i) lower coordination costs and a greater span of control and (ii) less reliance on specific information. We identify a class of mutual funds – balanced funds – in which the investment strategies predicted under centralized decision-making are different from those under decentralized decision-making. The empirical predictions are developed within a simple framework that helps to clarify the tradeoffs in centralized versus decentralized decision-making in market equilibrium. Our empirical results are strongly supportive of the decision-rights theory: investment strategies of sole-managed funds are less reliant on specific information; they reflect the manager’s greater span of control and lower coordination costs.

There are several distinct advantages from studying decision-making in organizations in the context of investment strategy and performance of balanced fund managers. First, US mutual

¹A large theoretical literature studies the implications of moral hazard problems in teams and the incentive schemes associated with the efficiency of teamwork (see, e.g., Holmström, 1982; Rasmusen, 1987; Prendergast, 1999).

funds are managed both by individual managers and by management teams. Management teams are typically composed of two or more managers that, depending on fund objectives, have expertise in different asset classes or industry sectors. The structure of fund management gives a straightforward way to categorize the decision-making structure – being more centralized under a sole-manager or less centralized when team-managed. Second, many decisions of fund managers, such as their portfolio holdings and trading activities are observable, which is not the case in most other settings. Furthermore, the performance of mutual funds can be reliably measured. The finance literature has developed a large set of performance evaluation methodologies that allow us to evaluate the performance of mutual funds and to characterize their investment strategies.

Finally, our focus is specifically on balanced funds because, as we will argue, they are particularly suited for testing decision-structure hypotheses. Balanced funds allocate investments across different asset classes, typically between stocks and bonds. They are usually required to maintain, with varying degrees of flexibility, a pre-specified ratio of debt and equity investments.² In broad terms, there are two types of investment strategies available to balanced funds. Balanced funds invest in both stocks and bonds and, hence, can gain performance through allocation decisions *across* asset classes (generally referred to as market timing skills) and/or by identifying investment opportunities *within* each asset class (referred to as security selection skills). While both types of strategies can contribute to fund performance, the structure of decision rights that facilitates one or the other strategy is different.

In team-managed balanced funds, individual managers are typically specialized in one of the asset classes.³ We should, therefore, expect security selection strategies to contribute to the performance of team-managed funds. A market-timing strategy, on the other hand, is less suited to a decentralized team structure since it places substantial coordination demands on the managers. Allocation decisions across asset classes will have to rely on the agreement

²The funds are often promoted as advantageous for investors seeking a simple way to achieve a broadly diversified holding of stocks and fixed-income investments. See, e.g., <https://advisors.vanguard.com/VGApp/iip/site/advisor/investments/mutualfunds/>.

³For instance, on 27th October 2011, the Eaton Vance Balanced Fund (A) website (<http://funds.eatonvance.com/Balanced-Fund-EVIFX.php>) listed its management team: Charles Gaffney is a portfolio manager on the “large-cap core/equity income” team while Thomas Luster and Bernard Scozzafava are portfolio managers on the “investment-grade fixed-income” team.

and coordination among the various managers specialized in particular asset classes. These coordination problems may be exacerbated by: conflicting incentives and information, the potential adverse impact of asset allocation decisions on security selection performance, and an unwillingness to reduce the assets under their control. Hence, a fund in which the decisions are made by a sole-manager faces fewer coordination problems. In such a centralized structure, the manager can unilaterally change allocations across asset classes and, therefore, has an inherent advantage in terms of implementing market-timing strategies. At the same time, a sole manager may not have the specialized skills necessary for successful security selection. Even if the manager has subordinates that specialize in particular asset classes, issues of agency and communication, highlighted in Jensen and Meckling (1992), will tend to deliver weaker performance than if the subordinates had independent decision rights over the asset classes and were evaluated on that basis.

For the empirical analysis, our sample of balanced funds includes two types of asset allocation funds from the Center for Research in Security Prices (CRSP) Survivor-Bias-Free US Mutual Fund Database: those that strive to maintain a balanced portfolio of stocks and bonds within a pre-specified range (“Rigid funds”) and those that are generally more free to allocate resources across the two asset classes (“Flexible funds”). We identify team- and sole-managed funds for the full sample of balanced funds based on the fund manager information provided in the same database.

We first study the asset allocation decisions, or market timing skills, in the sample of balanced funds based on fund returns. Results from the Treynor and Mazuy (1966) and the Henriksson and Merton (1981) market timing models reveal significant market timing in the full sample of balanced funds. However, there are systematic differences in the market timing performance between sole- and team-managed funds. In general, sole-managed balanced funds exhibit significant market timing skills while there is little evidence of market timing in team-managed funds. Further, sole-managed funds exhibit greater market timing performance in Flexible funds, where managers have greater discretion in their allocation decisions across asset classes, than in Rigid funds. To investigate further the relation between fund management structure and market timing decisions, we examine market timing activities in a sample of

funds that change their management structure, from team to sole or from sole to team, over the sample period. The results suggest a causal link between management structure and investment strategy: funds that change from sole management to team management experience significant decline in market timing activities; the converse holds for funds that change from team to sole management.

To better understand the impact of management structure on fund investment strategy and performance, we decompose the overall performance of funds into two components – that due to market timing and to security selection. We do this based on fund returns as well as asset allocation decisions. The results indicate that, while team-managed funds outperform sole-managed funds in security selection, they underperform in market timing, particularly among Flexible funds. As a result, the overall performance of the team- and sole-managed balanced funds is not significantly different, which is suggestive of a market equilibrium. The difference in investment strategy and source of performance between sole- and team-managed funds is consistent with the prediction that location of decision rights affects fund investment decisions.

We evaluate alternative explanations that are not based on decision rights for the lack of market timing in team-managed balanced funds. For instance, one possibility is that, for reasons other than decision-rights, fund companies assign managers with superior market timing skills to be sole managers of balanced funds, while using team-management when managers with specialized skills are recruited. To examine whether differences in fund objectives and/or managerial skills explain the lack of market timing in team-managed funds, we look for evidence of market-timing *within* the equity portfolios (i.e., shifting the risk levels of equity portfolios according to market conditions) in both sole- and team-managed funds. Using information from the equity holdings of these funds, we find robust evidence that sole- and team-managed funds are equally likely to engage in market timing within their equity portfolios. This suggests that the lack of market timing across asset classes in team-managed funds may not be due to a lack of such skills on the part of team managers that specialize in equities. Instead, it may largely be the result of communication and coordination costs within a decentralized structure.

While team-managed funds do worse in terms of market-timing, our evidence suggests that, because of the specialization of team members, they outperform sole-managed funds in security

selection within asset classes. We further study the relation between market timing and security selection within sole- and team-managed funds, and find a strongly negative correlation between performance due to market timing and security selection. The negative correlation within sole- and team-managed balanced funds suggests that market timing activity can, by itself, have an adverse impact on investment decisions associated with security selection. We also find that sole-managed funds typically hold far fewer stocks and more cash than team-managed funds, a finding that is further indicative of the costs associated with market timing.

We next investigate whether balanced fund investors respond differently to market timing performance depending on the management structure of the fund. Specifically, do they expect such strategies to be largely the domain of sole-managed funds? Examining the flow-to-performance relation, we find that investors do value market timing performance, but the positive relation between market timing performance and fund flows is largely driven by the sample of sole-managed funds. The lack of such a positive flow-to-performance relation for team-managed funds suggests that investors discount what appears to be market timing performance as an indicator of managerial skills and future performance. This may be because it is not the strategy that the investors expect the team-managed funds to be implementing.

We note that mutual fund companies may attempt to improve coordination within management teams by providing appropriate incentives to managers. For team-managed funds, for instance, fund companies may be able to promote coordination by rewarding managers for a fund's overall performance – in addition to rewarding their individual contributions. Further, to encourage market timing, fund companies could set performance evaluation benchmarks based on the fund's overall performance or based on explicit market timing benchmarks to improve overall fund performance.⁴ It is, however, an empirical issue as to how effective management incentives might be in mitigating coordination problems with teams. Our results suggest that, despite the possible use of incentives, coordination remains a substantial problem in funds.

Our paper is related to the literature on decision rights and organizational structure (Jensen and Meckling, 1992; Hart and Moore, 2005; Hart and Holmstrom, 2010) as well as to the

⁴See Prendergast (1999) for a review on incentive provisions within firms and van Binsbergen, Brandt, and Kojen (2008) for discussions relating to decentralized investment management.

literature on the economics of teams (Becker and Murphy, 1992). We conduct a direct test of the decision-rights theory in the specific context of balanced-funds. We show that the investment strategy and performance of balanced funds is consistent with the location of decision rights and the trade-off between specialization benefits versus coordination costs. As predicted by the hypothesis, team-managed (sole-managed) balanced funds reallocate investments less (more) frequently across asset classes but are more (less) specialized, as reflected in their security selection performance.

Our study also contributes to the mutual fund literature on market timing. While there exists a substantial theoretical literature on market timing (such as, Treynor and Mazuy, 1966; Merton, 1981; Henriksson and Merton, 1981; and Admati, et al. 1986), empirical studies on market timing typically focus on equity mutual funds and generally find little evidence of market timing.⁵ Our paper, however, studies mutual funds with investment objectives that encompass market timing decisions and finds significant market timing ability in this group of funds. More important, our analysis here reveals that various factors such as organizational and management structure can have a significant impact on whether fund managers pursue market timing strategies.

The rest of the paper is organized as follows. Section 2 presents a simple theoretical framework for deriving the testable predictions while Section 3 describes the balanced fund data. Section 4 introduces methodologies for evaluating market timing skills as well as provides evidence on the investment decisions and performance of team- versus sole-managed mutual funds. Section 5 explores the coordination costs in teams and also analyzes the trade-off between specialization benefits and coordination costs within the team structure. Section 6 concludes the paper.

2 A Simple Framework To Develop Testable Predictions

As we have discussed, decentralized decision-making provides benefits due to specialization, though at the cost of larger coordination costs. We illustrate the underlying economic rationale and obtain empirical predictions by developing a simple theoretical framework. We assume that

⁵For example, see Henriksson (1984), Ferson and Schadt (1996), and Becker et al. (1999).

a balanced fund has a total of two monetary units (say, two dollars for simplicity) available to invest among the two asset classes – bonds and equity. The fund is expected to invest one dollar in each asset class, though it is allowed to shift up to $\alpha < 1$ dollars between the classes. The investment in each asset class is assumed to be managed by a single specialist.

Funds can have one of two types of management structures. One is a team management structure, denoted by T , in which the specialists are given full decision rights to choose among investments in their respective asset class. It is assumed that, following a security-selection strategy, each specialist can independently put in effort and get a valuable investment idea with probability p_T that can produce an “excess” value of V for a 1 dollar investment. Here, “excess” value is the value in excess of that produced by an uninformed, passive investment in that asset class. With probability $(1 - p_T)$, the specialist is not successful at generating investment ideas and produces no excess value. Hence, the expected value from investments under a team-management structure is given by $\Pi_T = 2p_TV$.

The alternative management structure, denoted by S , is one in which there is a single manager with the authority to override the decisions of the specialists in the two asset classes. Suppose that this manager can generate information with a probability p_M about which of the two asset classes will perform better than the other. He can reallocate up to α dollars to the class that he expects will perform better. If the per-dollar difference in performance is A , then the manager can contribute a value of αA through the reallocation. When the manager has no special information, he lets the specialists make their own investment decisions. We allow for the possibility that, on account of the likelihood of asset reallocation (and other costs of reallocation), there is a negative incentive effect on the specialist asset managers such that the probability with which they generate ideas is $p_S \leq p_T$. Hence, the expected value of investments under the sole-management structure is:

$$\Pi_M = p_M(\alpha A) + 2p_S V.$$

If both types of structures exist in the market, this would imply that the returns generated under the two structures are equal in equilibrium, i.e., we would expect $\Pi_M = \Pi_T$. If there is heterogeneity in terms of skills in stock selection and market timing, we would expect fund

sizes and fees to adjust such that investors are indifferent between investing in one type of management structure versus the other.⁶

Despite its simplicity, the above discussion generates several testable implications that we describe below and that are investigated in our empirical tests.

1. The sole-management structure in balanced funds is more likely to be associated with market timing investment strategies while the team-management structure is more likely to encourage security selection strategies.
2. A sole-managed balanced fund is more likely to switch allocations across different asset classes than a team-managed balanced fund.
3. The average performance of the two types of management structures (sole and team) are expected to generate a similar performance on average, even though the returns may be realized from different strategies (market-timing or security-selection).

There are other issues that we pursue in our empirical analysis as well. We investigate whether there appear to be inherent differences in the market-timing ability of sole and team-managed funds by investigating evidence of market-timing within the fund's equity portfolios (rather than across asset classes). We also test whether the flow-to-performance sensitivity of funds corresponds to the strategy that investors expect the managers to be following. Hence, for sole-managed funds, given that investors expect them to focus on market-timing, the flows may respond largely to market-timing performance instead of returns that are ostensibly due to security-selection.

3 Data and Sample Description

3.1 The Balanced Fund Sample

The data used in this study are from the CRSP Survivor-Bias-Free US Open-End Mutual Fund Database. Our sample period is from 1992 to 2009, since fund-manager information becomes

⁶In principle, while we do not pursue this, the equilibrium could be extended to accommodate various types of heterogeneity in the efficiency and costs of fund operation as well as differences in the types of managers that are drawn to the alternative structures.

available in the database only after 1992. From this database, we select all asset allocation funds that are designated as “Balanced Funds” or “Flexible Portfolio Funds”, i.e., funds that have Strategic Insight (SI) objective code of “BAL” or “FLX”, or have Lipper objective code of “B” or “FX”.⁷

Both the “Balanced funds” (“BAL” or “B”) and “Flexible portfolio funds” (“FLX” or “FX”) belong to the broad category of asset allocation funds and are generally referred to as balanced funds in the industry. While both types allocate assets between stocks and bonds, there are important differences: “Balanced funds” typically do not deviate substantially from the fixed proportions specified in the investment objective of the funds (for example, 60% in equity and 40% in bonds); “Flexible portfolio funds”, on the other hand, are often free to allocate up to 100% of the funds to one asset class. Throughout the paper, we refer to both types of funds as balanced funds, and designate the first type as “Rigid” and the second as “Flexible”.

All funds in the sample are classified as being “sole” or “team” managed, based on their management structure. For funds managed by a single manager, the manager is reported by name in the CRSP mutual fund database. We designate these funds as sole-managed funds. If CRSP lists the names of two or more managers, or uses the terms: “team”, “management team”, or “investment committee” to describe the managers of the funds, then we label these funds as team-managed. We keep track of a fund’s management structure for each year as it can change over time. Because the CRSP mutual fund database treats multiple classes of a single mutual fund as separate funds, we combine the different classes into one single fund to obtain fund-level variables, such as expense ratio and returns. These variables are weighted by the total net assets (TNA) of each class within the fund at the beginning of every quarter. Our final balanced fund sample includes funds that have appeared in the CRSP database for at least three years.

3.2 Summary Statistics

Table 1 reports the descriptive statistics of the sole and team-managed balanced funds over the sample period. As indicated, the total number of balanced funds increased in the first half

⁷When there is a difference between the Lipper classification and Strategic Insight classification, we use the Lipper classification because Lipper’s classification is more recent and more detailed.

of the sample period and then remained largely stable, with a slight decline towards the end. There is also a clear decline in the share of sole-managed funds over this period, consistent with a similar evolution in the mutual fund industry (see, e.g., Massa, Reuter and Zitzewitz, 2010), though the trend is apparently stronger for balanced funds than, say, equity funds in general (Bäer, Kempf and Ruenzi, 2011).

Sole-managed funds are more likely to be Flexible funds, with about seventy percent of Flexible and half of Rigid funds being sole-managed in 1992. While the fraction of sole-managed funds has declined, they still account for about thirty percent of Flexible funds in 2009, though only twenty percent of Rigid funds. The fraction of Rigid funds has increased over the sample period, as has the fraction of Rigid funds that are team-managed. The average fund size has increased over the sample period.

Table 2 reports fund characteristics for the sole- and team-managed funds. The table reports the time series average of mean monthly returns, their standard deviations, fund expense ratio and the annual turnover ratio for Flexible and Rigid funds separately. The mutual fund returns are net of fund expenses. For the full sample, as well as for the Flexible and Rigid subsamples, univariate tests indicate that the difference in returns between sole and team-managed funds is not statistically significant. A noteworthy difference is that sole-managed funds have expense ratios that are 10 basis points higher than those of team-managed funds. Also, Flexible funds have expense ratios 20 basis points higher than those of Rigid funds.

Table 3 examines the asset allocation policies of balanced funds. We use year-end information on asset allocation among the three asset classes: stock, bond and cash.⁸ For each year, we obtain average asset allocations and compute the cross-sectional dispersion of asset allocation among the funds. On average, balanced fund managers in the sample invest about 60% of their assets in stocks, 30% in bonds, and 10% in cash. There is, however, considerable dispersion across funds in their asset allocation policies. For instance, sole-managed funds typically allocate a greater portion of their assets to stocks – and less to bonds – than team-managed funds. Cash constitutes about 11% of the portfolio value in the full sample. Among Flexible funds,

⁸Bond investments include corporate bonds, treasuries, municipal bonds, and convertible bonds. Prior to 1999, CRSP provides yearly updates of asset allocation. Starting from 1999, quarterly updates of portfolio allocation across asset classes became available.

sole-managed (team-managed) ones hold 15% (12%) of their assets as cash.

In the next section, we employ various mutual fund performance evaluation models to examine the asset allocation decisions and their contribution to fund performance.

4 Evidence of Market Timing in Sole- and Team-Managed Funds

Both asset allocation or, equivalently, market-timing (moving investments across asset classes) and security selection (investing in securities within an asset class) can contribute to a fund's overall performance. In this section, we employ standard approaches to evaluate the market timing ability and performance of balanced funds. In particular, we test whether management structure affects the use of market timing strategies in the manner predicted.

4.1 Measuring Market Timing of Balanced Funds

There are two traditional models that have been used to evaluate the market timing ability of mutual funds: the Treynor and Mazuy ("TM") model, and the Henriksson and Merton ("HM") model. For fund i , the TM market timing measure is the estimated $\gamma_{i,TM}$ in the following model based on fund i 's returns:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,m}(r_{m,t} - r_{f,t}) + \gamma_{i,TM}(r_{m,t} - r_{f,t})^2 + \epsilon_{i,t}. \quad (1)$$

The HM market timing measure is the estimated $\gamma_{i,HM}$ from the following model:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,m}(r_{m,t} - r_{f,t}) + \gamma_{i,HM}[r_{m,t} - r_{f,t}]^+ + \epsilon_{i,t}. \quad (2)$$

In the above models, $r_{i,t}$ and $r_{m,t}$ are the time t returns for fund i and the value weighted stock market, respectively, while $r_{f,t}$ is the return on risk free asset (proxied by the 30-day U.S. T-bill) in month t . Both models are intended to capture convexity in the relationship between the stock market and fund returns. In the TM model (Treynor and Mazuy, 1966) market timing activity is captured by the squared term in the model. Hence, a positive value of γ in (1) indicates market timing ability. In the HM model (Merton and Henriksson, 1981), convexity is modeled in the form of a call option payoff: $[r_{m,t} - r_{f,t}]^+$ (or, equivalently, $\max(r_{m,t} - r_{f,t}, 0)$): A successful market timer is, in effect, producing free, or "cheap" call options on the market. Again, a positive value of γ in (2) implies market timing ability.

Both the TM and HM models focus on the market timing ability of equity mutual funds that switch between equity and the safe assets (T-bills). However, as argued in Elton et al. (1993) and Blake, Elton, and Gruber (1993), balanced funds invest a substantial portion of their assets in bonds and, hence, a performance evaluation model should include a bond index, in addition to equity indices. Also, tests for market timing should capture shifts between stocks and bonds (see also Aragon, 2005, and Commer, 2004). We thus enhance the TM and HM market timing models as follows:

The two-factor TM model:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,m}(r_{m,t} - r_{f,t}) + \beta_{i,b}(r_{b,t} - r_{f,t}) + \gamma_{i,TM}(r_{m,t} - r_{b,t})^2 + \epsilon_{i,t}. \quad (3)$$

Similarly, the two-factor HM model:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,m}(r_{m,t} - r_{f,t}) + \beta_{i,b}(r_{b,t} - r_{f,t}) + \gamma_{i,HM}[r_{m,t} - r_{b,t}]^+ + \epsilon_{i,t}. \quad (4)$$

In the above models, the additional factor, $r_{b,t}$, is the return on a bond index; the market timing factor also includes $r_{b,t}$. The coefficient γ_i provides a measure of market timing though, unlike traditional models (1) & (2), it represents shifts between the equity and bond markets. Since models (3) & (4) include both stock and bond indices, we will refer to them as two-factor models throughout the paper.⁹ To control for additional risk factors in evaluating equity portfolio returns, we conduct tests in which we control for the size, value and momentum factors (Fama and French 1993, Carhart 1997). We refer to these models as five-factor models since they include the three additional risk factors. These models serve as the baseline models in our market timing tests. In all empirical tests, $r_{m,t}$ is the CRSP value weighted stock market return, $r_{f,t}$ is the return on the one-month Treasury-bill, and $r_{b,t}$ is the total return on the Barclays Capital aggregate bond index (previously Lehman Brothers aggregate government/corporate bond index). The three factors are the returns on the zero-investment factor-mimicking portfolios for size, book-to-market, and one-year stock return momentum.

⁹Here we only consider the market timing activity between the equity market and bond market. In various robustness tests, we also consider market timing between bond portfolios and risk free assets by introducing an extra term of $[r_{b,t} - r_{f,t}]^+$ or $(r_{b,t} - r_{f,t})^2$ (see Aragon, 2005). The results show that the bond timing coefficients are generally insignificantly different from zero. We leave these results unreported.

4.2 Evidence from Return-based Market Timing Measures

The main results on management structure and investment strategy are presented in this subsection. We assess the impact of fund management structure on market timing and overall fund performance. Further, we investigate changes in fund investment strategy following changes in their management structure over the sample period.

4.2.1 Full Sample Results

We begin by estimating the market timing measures based on the two- and five-factor models. For each balanced fund in the sample we estimate the models using monthly returns over a three-year rolling window. At the end of each year, we compute the average of the market timing measure estimated for the team- and sole-managed balanced fund portfolios, respectively, and further for the sub-samples of Rigid and Flexible funds. We then compute the time series means of the average market timing coefficient estimates over the sample period and compute the t -statistics for the time series mean.

Table 4 reports the results for the full sample of balanced funds and separately for sole- and team-managed funds. Panel A reports the results based on the Treynor-Mazuy model. For the full sample, the average market timing coefficient based on the two-factor model is positive, but not significantly different from zero. The five-factor model that includes the *Size*, *Value*, and *Momentum* factors provides stronger evidence of market timing with a statistically significant average market timing coefficient. For the sub-samples of Rigid and Flexible funds, the market timing measures are found to be significantly positive in both sub-samples, though Flexible funds appear to have a greater ability to time the market, as indicated by the magnitude of the estimated coefficients.

Next, we examine the market timing skill of sole- and team-managed funds separately. Since funds change their management structure over the sample period – switching team to sole or the other way – we compute average market timing ability based on funds that do not change their structure over the three-year period in which performance is evaluated. The results indicate that sole-managed funds exhibit significant market timing ability, particularly based on the five-factor models. In contrast, team-managed funds do not, on average, exhibit market timing

ability. The market timing ability observed in the overall balanced fund sample is, therefore, largely driven by the sole-managed funds in the sample. Further examination reveals that among sole-managed funds, it is the Flexible funds that exhibit greater market timing abilities. There is no significant evidence of market timing in the Flexible and Rigid sub-samples of team managed funds.

Panel B provides results based on the Henriksson-Merton model. The results are largely consistent with the ones from the Treynor-Mazuy model: we observe market timing ability in the full sample of balanced funds, but such evidence is largely driven by the sole-managed funds. There are some minor differences between the results from the HM and TM models. First, for the full sample results, the Rigid fund sub-sample does not show significant market timing ability, for both the two- and five-factor model. Second, related to the full sample results, the sole-managed Rigid funds do not show significant market timing ability.

A few observations are in order. First, for the full sample of balanced funds, we find significant market timing ability based on the five-factor market timing models while the two-factor model results are generally weaker. Because the fund managers' decision to shift between the equity and bond portfolios could be partly driven by their forecasts of the time varying premiums of the risk factors, i.e., the *Size*, *Value*, and *Momentum* factors, controlling for these risk factors yields more robust tests of the market timing ability. Second, there are systematic differences in the observed market timing ability between sole- and team-managed funds. In general, sole-managed balanced funds exhibit significant market timing ability while team-managed funds don't. Third, for sole-managed funds, market timing ability is more evident in the Flexible funds compared with the Rigid funds. The results are consistent with the notion that Flexible fund managers have greater discretion in allocating investments across asset classes. Notably, there is little difference in the market timing ability between the team-managed Rigid funds and Flexible funds as neither shows significant market timing ability.

4.2.2 Management Structure Changes and Market Timing

We now present further evidence on the impact of fund management structure on fund market timing ability. Over the sample period, a number of balanced funds went through a change

in management structure. Part of the reason for these changes is the gradual shift over time from sole to team management in the mutual fund industry, including the balanced fund sector (Table 1). At the same time, some balanced funds switched the other way: from team to sole management. In this subsection, we study the sample of balanced funds that changed structure and examine whether performance and market timing is affected by such changes. Because sample sizes within Rigid and Flexible funds are small, we do not separately examine Rigid and Flexible funds within the management-change sample.

The management-change sample consists of funds that experience changes from team to sole management and from sole to team management. Furthermore, we require that the funds have at least three-year return information before and after the year of the structural change. In total, 95 funds changed from sole to team management and 22 funds changed from team management to sole management. We employ the market timing models and use the three-year returns before and after the year of management change to evaluate fund performance and its market timing abilities. For comparison, we also include one performance evaluation model without the market timing factor.

Table 5 reports the average α 's and the average market timing coefficients before and after the management-change and the differences between the “before” and “after” periods for the same funds. The results are obtained using three different models: the five-factor model without market timing, the five-factor TM, and five-factor MH market timing models. Panel A reports results for funds that changed from sole management to team management. In the standard performance evaluation model without market timing, the α s before and after the change are both significantly negative, with the α declining by a statistically insignificant amount for the funds that moved from sole to team management. The negative α s here are not surprising because the fund returns are net of the expense ratio. From the Treynor-Mazuy and the Merton-Henriksson market timing models, we obtained the market timing coefficients for both the before- and after-change periods. It is striking that the sole-managed funds exhibit significant market timing ability before the funds change to team management, but exhibit no market timing ability after the change. The difference of the before-after market timing coefficients are also significant. Additionally, there is a noticeable decline in the estimated α 's after the change

based on the market timing models. As we discuss in the next subsection, the α 's from the market timing models can be interpreted as a proxy for security selection skills. The results suggest that fund performance – from security selection or market timing – appears to worsen after the shift from sole to team management.

The converse is generally true for the sub-sample of funds that changed from team- to sole-management. There is significant improvement in the estimated α s from the standard performance evaluation models as well as from the market timing models. What is most compelling is that prior to the change the team-managed funds do not exhibit significant market timing but do exhibit significant market timing ability after the change. Notice that the improvement of fund performance, α , is largely driven by the underperformance of the team managed funds before the structure change, rather than the outperformance of the funds after the change. The annualized α s from the market timing models are -5.8% and -8.0% respectively based on the Treynor-Mazuy and the Merton-Henriksson models, much lower than the overall sample results (see Table 6 in the next subsection). As noted earlier, the general trend in the mutual fund industry, as well as in balanced funds, is a shift from sole management to team management. Hence, the change from team- to sole-management may have been triggered by the significant underperformance of the team-managed funds. However, what is key for our purpose is that the change from team to sole management has considerable impact on market timing strategy.

To briefly summarize our findings: full sample results show that fund management structure is significantly related to the market timing ability of balanced funds. Results based on funds that experience management structural changes suggest that the relation may be causal: fund investment strategy is determined by management structure adopted by the funds.

4.2.3 Market Timing, Security Selection and Fund Performance

In this subsection, we assess the relative contribution of market timing and security selection to the overall performance of balanced funds. Our focus is on the effects of management structure on security selection and the overall fund performance.

We obtain α s and the market timing coefficients from regression results based on the Treynor-Mazuy and the Merton-Henriksson models using the procedure that we described previ-

ously. In standard performance evaluation models without market timing, the α is the abnormal performance of a fund after controlling for risk factors and is usually interpreted as measuring overall performance or managerial security selection skills. In evaluation models that also measure market timing, the α from the regression model can be interpreted as “timing-adjusted” stock selectivity. It should be kept in mind, however, that stock-selection and market-timing strategies are not necessarily independent. For instance, a fund’s market timing actions can affect security selection if, say, the fund is seeking to liquidate equity positions to decrease stock market exposure. In such cases, α provides a measure of security selection ability that has been distorted by market-timing decisions. Similarly, market timing measures may also reflect the impact of security selection. The fact that, in general, it is difficult to completely disentangle the two investment strategies is well recognized in the literature (see for example, Glosten and Jagannathan, 1994; Ferson and Qian, 2004).¹⁰

Table 6 presents results on fund performance based on the five-factor market timing models. We first obtain the α ’s from the regressions for each fund and for each year based on the three-year rolling window. We then compute the contribution of market timing to the fund’s performance, using the estimated γ coefficient and the market timing variables. Finally, we obtain the sum of α ’s and the returns attributed to market timing for each fund: with the sum providing a measure of the fund’s overall performance. With the afore-discussed caveats on the two performance measures, we will use security selection (SS) and market timing (MT) to refer to the two components of fund performance. We first compute the average values of the three variables (SS, MT & Overall) for each year and then obtain the time series mean of the average values and the associated t -statistics. The table reports results based on before-fee returns (Panel A) and net-of-fee returns (Panel B).

As indicated in Table 2, expense ratios differ between sole- and team-managed funds (sole-managed funds have higher expense ratios) and between Rigid and Flexible funds (Flexible funds have higher expense ratios). We first present the before-fee results because these managerial

¹⁰Because our emphasis in the analysis is the comparison between sole and team managed funds, these concerns are mitigated. We estimate performance evaluation models with and without market timing and find consistent results for overall fund performance. In the next section, we disentangle the market timing and security selection based on portfolio weights in equities and bonds. In various different tests, we find consistent evidence across different performance evaluation models.

skill (performance) measures are not affected by the different expense ratios across the types of funds. For the before-fee results, we add back the annual expense ratio to monthly returns and run all the market timing models based on gross returns. Both the α and market timing performance are annualized in the reported results. Because the results based on the Treynor-Mazuy model and the Henriksson-Merton model are similar, we base our discussions on the the Treynor-Mazuy model unless noted otherwise.

Starting with the full sample, the results show that balanced funds generate returns that are indistinguishable from their respective benchmarks. It is worth noting that, for the full sample, Flexible funds seem to outperform Rigid funds, as indicated by the magnitudes of the overall performance of the two types of funds. In fact, Rigid funds under-perform the risk benchmarks, while Flexible funds do not show abnormal performance. This difference in performance is less evident based on the net returns since Flexible funds tend to have a higher expense ratio than Rigid funds.

We next examine the security selection and the market timing components of fund performance in the sole- versus team-managed funds. Starting from the security selection component, only team-managed Flexible funds exhibit some security selection ability. As measured by the differences between the sole- and team-managed funds, there is little difference in the security selection performance between the two groups in Rigid funds, but team-managed Flexible funds significantly outperform their sole-managed counterpart in security selection (50 basis points annualized). The results based on the Henriksson-Merton model are consistent with those from the Treynor-Mazuy model.

The performance contribution of market timing in the overall sample is negative, though this is only significant for the Rigid fund sample.¹¹ Market timing contributes positively to sole-managed funds, particularly Flexible funds. However, the performance contribution of market timing for team-managed funds is significantly negative for Flexible funds. As a result, sole-managed funds outperform the team-managed funds in market timing related performance, in both Flexible funds and Rigid funds. Results from the Henriksson-Merton model are similar

¹¹The average performance attributable to market timing is different from the average value of the market timing coefficient. The negative performance is partly caused by the incorrect market timing when returns between equities and bonds diverge greatly.

for sole-managed funds, but show significantly negative market timing contribution for team-managed Rigid funds and marginally negative contribution for Flexible funds. As discussed previously, the “market timing” components here can be affected by security selection strategies, but the comparison between the two groups provides preliminary results on the differences in the market timing performance of sole- and team-managed funds.

Aggregating the investment performance from both security selection and market timing, we find that the sole-managed Rigid funds outperform team-managed Rigid funds and sole-managers slightly underperform team-managers in Flexible funds. The Flexible fund results reflect the fact that, while sole-managed funds outperform team managed funds in market timing, they underperform in security selection.

We now briefly discuss the results based on fund net returns. Because the difference between net return results and gross return results largely reflects the impact of the expense ratio, the comparison between sole and team managed funds is similar. Two differences are worth noting. First, the security selection component and, consequently, the overall performance of the funds is significantly negative because of the deduction of the expense ratio from the gross returns. Second, because sole-managed funds tend to have higher expense ratios, the overall performance difference between sole- and team-managed funds changes somewhat from the gross return results: the outperformance of sole-managed Rigid funds is lower, and the underperformance of the sole-managed Flexible funds is greater.

4.2.4 Further Robustness Tests

We conduct several robustness tests on the market timing and fund performance results. One robustness test considers market timing between bond portfolios and risk free assets by introducing an extra term $[r_{b,t} - r_{f,t}]^+$ or $(r_{b,t} - r_{f,t})^2$ in the market timing models (see Aragon, 2005). The results show that the bond timing coefficients are generally insignificant from zero and the timing coefficients based on the relative returns of equity and bond portfolios are similar to those reported in the tables. In another robustness test, we adopt the conditional performance evaluation framework to examine fund performance and market timing abilities (see Ferson and Shadt, 1996). We include the conditioning variables in the conditional market

timing models. The results show that fund managers incorporate public information in market timing decisions and the market timing results of fund managers are partially explained by the public information. We also examine evidence of market timing in three six-year sub-periods in our sample. We find stronger evidence of market timing in the first period than in the last period; this is possibly due to the greater prevalence of sole-managed funds in the first period of our sample.

5 Decentralization & Coordination Costs in Fund Management

In this section, we first evaluate an alternative explanation for our findings: that the relation between management structure and investment strategy might be the result of sole and team fund managers not sharing the same investment objectives or having different skills in market timing. We then explore a variety of issues related to the tradeoff between specialization benefits and coordination costs, its relation to fund management structure decisions, as well as investor flow response to different components of return.

5.1 Market Timing within Equity Portfolios

We address the possibility that our results on the relation between investment strategy and management structure are the result of managerial investment styles and abilities, rather than the structure of decision rights, as we have argued. For example, fund families may (for unspecified reasons) assign managers with good market timing skills to manage funds as sole managers and assign managers with specialized security selection skills (within equities or bonds) to manage the equity or bond portions of fund portfolios.

To test the alternative explanation, we investigate market timing within the equity portfolios of the balanced funds. The notion is that fund managers can substantially alter the exposure of their equity portfolios to the stock market, based on their forecasts of equity returns. If sole- and team-managed funds have different investment objectives or skills with regard to market timing, we would expect team-managed funds to not engage in market timing trades: whether *across* asset classes or *within* their equity portfolios. However, if it is the team structure and associated coordination costs that hinder reallocation across asset classes, then we might expect

fund managers to actively pursue market timing strategy within the equity portfolio – while having difficulty implementing it across the asset classes.

Information on the returns of the equity portfolio of a balanced fund is not directly available. We thus examine market timing within the equity portfolio based on the equity holdings of the balanced funds. In fact, Jiang, Yao and Yu (2007) argue that portfolio holding based market timing measure is less subject to the “artificial timing” bias present in security returns and can provide a good alternative measure on market timing.

The holdings-based market timing approach involves estimating the fund beta on the basis of its portfolio holdings – and then testing the fund beta for evidence of market-timing, using the Treynor-Mazuy and the Henriksson-Merton models. We first estimate the fund beta as the weighted average of the beta estimates for all the stocks held by the fund, that is,

$$\hat{\beta}_{i,t} = \sum_{i=1}^N \omega_{i,t} \hat{b}_{i,t}$$

where $\omega_{i,t}$ is the portfolio weight for stock i at the beginning of the holding period t , and $\hat{b}_{i,t}$ is the beta for stock i estimated using data prior to period t . Following the Treynor-Mazuy and the Henriksson-Merton models, we measure market timing by estimating the coefficients γ from the following models:

$$\beta_{i,t} = \alpha_i + \gamma_{i,TM}(r_{m,t} - r_{f,t})^2 + \eta_{i,t}, \quad (5)$$

$$\beta_{i,t} = \alpha_i + \gamma_{i,HM}[r_{m,t} - r_{f,t}]^+ + \eta_{i,t}. \quad (6)$$

We refer to the γ coefficients estimated from (5) and (6) as the holdings-based Treynor-Mazuy measure and the holdings-based Henriksson-Merton measure, respectively. We obtain the equity holdings of the balanced funds from the Thomson-Reuters mutual fund holding database and estimate the fund beta for each quarter. The regressions are estimated for each fund over the full sample period, so long as there is no change in the management structure over the period. If the management structure of a fund changes, we run separate regressions for each of the two (or more) periods over which the structure is unchanged and treat the estimates as independent observations. Table 7 reports the average market timing coefficients based on models (5) and (6) for sole and team-managed funds in both Rigid and Flexible fund categories.

The results provide clear evidence of market timing activity for the sample of balanced funds. Importantly, there is little difference between the sole- and team-managed funds in terms of the observed market timing skills within equity portfolios. Based on the Treynor-Mazuy measure, sole- and team-managed funds, in both the Rigid and Flexible categories, exhibit significant market timing activity. The Henriksson-Merton measure provide similar, though slightly weaker results. Again, there is virtually no difference between the team- and sole-managed funds. These findings indicate that it is unlikely that there are substantial differences in investment objectives or market-timing skills between team- and sole-managed fund managers.

The empirical findings support the notion that there exist material coordination costs in team-managed funds. As we have argued, coordination costs in team-managed funds would be expected to adversely affect market timing across asset-classes – but would have little impact within an asset class, since the asset classes are typically controlled by individual managers. The results also suggest that market timing tests that focus only on shifts between equity and bond portfolios can underestimate the true overall market timing ability as both sole- and team-managed fund managers engage in market timing trades within their equity portfolios independent of (or in addition to) market timing across asset classes.

5.2 The Costs of Market Timing and Fund Management Structure

While team-management funds face greater coordination costs, they benefit from manager specialization. The evidence that team-managed funds outperform sole-managed funds in terms of security selection is indicative of such specialization benefits. In this sub-section we investigate whether stock selection tends to suffer on account of market-timing activities that may impose externalities on the stock selection process. Given that one major investment objective of the balanced funds is to allocate investments across different asset classes, the analysis can shed light on the question: why fund families adopt team management structure in the presence of significant coordination costs.

Trading associated with market timing (not coordination costs *per se*) can have an adverse impact on fund performance. First, a market timing strategy based on shifts between equity and bond portfolios can incur considerable costs. In addition to direct trading costs, substantial

shifts between equity and bond portfolios can incur price-impact costs, particularly in securities with low liquidity. Such costs can dilute the contribution of market timing trades to fund performance even when the fund manager has market-timing skills. Second, market timing objectives can interfere with security selection related investment strategies *ex ante*. For example, because of anticipated market timing trading needs, fund managers could emphasize the liquidity of the securities in their portfolio and limit their investment choices. Such constraints could also limit the use of valuable security selection related information by fund managers. In what follows, we assess the costs of market timing by examining the relation between market timing and security selection, and then present more direct evidence on the potential costs of market timing.

5.2.1 Relation between Market Timing and Security Selection

We first study the relation between market timing and security selection *within* funds with the same management structure. Examining such a relation within the two groups of funds helps to establish the impact of market timing on security selection and vice versa – without possible contamination resulting from differences in fund investment strategies due to the management structure of the funds.

We use the two variables from the regression results of the market timing models: the “timing-adjusted” security selection skills and the market timing component of fund performance. As noted earlier, the two variables cannot measure the security selection and market timing ability perfectly. To ensure the robustness of the results, we also implement a performance attribution procedure to decompose the overall fund performance into security selection and market timing. The performance attribution procedure is as follows.

The difference between the performance of a balanced fund and the performance of a benchmark portfolio can be defined as:

$$R_{P,t} - R_{B,t} = \sum_{i=1}^K w_{Pi,t} R_{Pi,t} - \sum_{i=1}^K w_{Bi,t} R_{Bi,t}. \quad (7)$$

$R_{P,t}$ and $R_{B,t}$ are the returns of the balanced fund and the benchmark portfolio in time period t respectively. We can assume that we have three asset classes ($K = 3$) for the balanced funds

and for the benchmark portfolio: stock, bond and cash. In the above equation, $w_{P,i,t}$ is the fund holding weight in asset class i , and $R_{P,i,t}$ is the return from investing in asset i over the evaluation period t by the fund. $w_{B,i,t}$ is the weight of the benchmark in asset class i , $R_{B,i,t}$ is the return on the benchmark index over the evaluation period t .

The difference in returns between the managed portfolio, P , and the benchmark portfolio, B , represents the total contribution from the managerial skills. We can further decompose the expression $(\sum_{i=1}^K w_{P,i,t} R_{P,i,t} - \sum_{i=1}^K w_{B,i,t} R_{B,i,t})$ into two terms:

$$\sum_{i=1}^K w_{P,i,t} R_{P,i,t} - \sum_{i=1}^K w_{B,i,t} R_{B,i,t} = \sum_{i=1}^K (w_{P,i,t} - w_{B,i,t}) R_{B,i,t} + \sum_{i=1}^K w_{P,i,t} (R_{P,i,t} - R_{B,i,t}). \quad (8)$$

Note that the first term on the right hand side captures the return difference between the fund's weights in the different asset classes and the benchmark weights, and is referred to as representing the "market timing" skills of fund managers. The second term represents the return differences between the actively managed portfolios and the benchmark indexes, and is labeled as representing the "security selection" skills of fund managers.

For the benchmark indexes, we use the CRSP value weight stock index, the Barclays Capital aggregate bond index, and one-month T-bill respectively. For the benchmark weights of the three asset classes, we compute the average weight for Rigid and Flexible funds for each quarter. Because we do not directly observe the returns of the three asset classes separately in a balanced fund portfolio, we can not compute the "security selectivity" skills directly. We obtain the total contribution of fund manager skills from the left hand side of equation (8) based on the overall before-fee fund returns and the benchmarks portfolio returns based on our constructed benchmark weight and the index returns.¹² We measure the market timing skills using the first term on the right hand side of equation (8) based on the observed portfolio weights of each fund. From equation (8), the difference between the overall excess return and the market timing skills is the "security selectivity" skills. While the performance attribution approach is widely used in both academic research and in the industry, one drawback of this approach is that the performance measures are not risk-adjusted.

In the empirical analysis, we use the market timing and security selection measures obtained

¹²We add back the expense ratios to fund returns to measure the differences between fund returns and benchmark portfolio returns because the index returns are gross returns.

through the performance attribution procedure above, as well as the regression based measures from the Treynor-Mazuy and Henriksson-Merton models. To examine the relation between market timing and security selection skills, and the relation between market timing skills and the overall fund performance, we run cross-sectional regressions for the full sample of balanced funds as well as within the sole- and team-managed sub-samples. We run the regressions for each year, and report the time series averages of the regression coefficients. We include several control variables, such as fund size (logarithm of total net assets), expense ratio, turnover, and fund age in the cross-sectional regression.

Table 8 reports the results. Panel A examines the relation between market timing skill and security selection skills. We calculate the market timing skill and security selection measures from: the performance attribution approach in Model 1, Treynor-Mazuy model in Model 2, and the Henriksson-Merton model in Model 3. From the full sample results, it is clear that there is a significantly negative relation between market timing skills and security selection skills across funds. The full sample result is not surprising, given our previous evidence that sole- and team-managed funds perform differently on market timing and security selection.

We next examine the relation between market timing skills and security selection skills separately within sole- and team-managed funds. We again find a significantly negative relation between market timing skills and security selection skills. The relation holds for both sole- and team-managed funds, and it also holds across both Rigid and Flexible categories. Note that all three sets of market timing and security selection measures yield similar results. Other than the coefficients on the expense ratio, all other variables have the same signs and similar levels of significance. Because Model 1 uses gross returns based on the performance attribution approach, there is no significant relation between fund expense ratio and fund performance, while the regression measures based on net returns show a significantly negative relation between expense ratio and the security selection skills.

We now briefly discuss the relation between market timing skills and overall fund performance. Throughout our empirical analysis, overall fund performance is the sum of the market timing component and the security selection component. So in the performance regression, the coefficient on the market timing variable is one plus the coefficient from the security selection

regression while all the other coefficients stay the same. Panel A shows that market timing skills and security selection skills are negatively correlated, but the magnitude of such impact is much smaller than 1. The regression coefficients indicate that while market timing negatively relates to security selection, the performance generated from market timing skills is not fully offset by the decline in the performance attributed to security selection skills. Results in Panel B confirm that this is indeed the case: market timing skills are positively related to overall fund performance.¹³

To summarize, there is a strong negative relation between market timing and security selection skills, in the full sample as well as within the sub-samples of sole- and team-managed funds. The evidence indicates that market timing activities can have negative impact on security selection performance, thus supporting the notion that market timing efforts limit security selection. However, the evidence also clearly shows that the negative impact of market timing on security selection is much lower than its positive contribution to overall fund performance. As such, the adverse impact of market timing on security selection should play a role in the management structure decisions of the mutual fund companies, but does not completely explain the choice of fund management structure.

5.2.2 Holdings and Liquidity of the Equity Portfolio

We now provide some more direct evidence on the possible channels through which fund management structure and market timing trading affect overall fund investment decisions. We first examine how sole- and team-managed funds differ in the number of stocks held in the equity portfolios. For each fund, we count the number of stocks held by the fund at the end of each year, and compute the mean (and median) number of stocks held by sole- and team-managed Rigid and Flexible funds. Panel A in Table 9 reports the averages of the mean and median values over the sample period. As shown in the table, team managed funds hold a larger number of stocks, as measured by both the mean and median number of stocks, and the results hold for both Rigid and Flexible funds. Panel B reveals that the differences in the number of stocks are not driven by the size differences of the equity portfolios (or size of the overall fund): except

¹³Coefficients on the control variables are not reported in Panel B because they are the same as in Panel A.

for the mean value of equity holdings in Rigid funds, sole-managed funds are larger than or similar to team managed funds in terms of fund size. There are several non-mutually exclusive explanations for the different number of stocks held by sole- and team-managed funds. Team members managing the equity portfolios could devote more resources on equity research and expand their research coverage on equities. Since team-managed engage less in market timing and are better at security selection, they can invest in more stocks of their choice.

Cross-asset market timing trades require greater liquidity within asset portfolios. Therefore, we next evaluate the liquidity of the stock portfolios of the balanced funds. We compute the Amihud (2002) (il)liquidity measure of each stock using daily returns and trading volume within one quarter, and then compute value-weighted and equally-weighted liquidity measures for each fund. Panel C shows mixed results on the comparisons between sole- and team-managed funds. Based on value-weighted results, sole-managed funds hold stocks with lower liquidity, particularly among Rigid funds. But the pattern reverses when comparing equally-weighted liquidity measures for Rigid funds. As shown in Panel A, team-managed funds invest in more stocks; so, the lower equally-weighted portfolio liquidity suggests that team-managed funds invest in a greater number of stocks with lower liquidity. The liquidity of sole- and team-managed Flexible funds is not significantly different.

The results on liquidity of equity holdings could signal that fund managers do not manage liquidity needs *within* the equity portfolio (see also, Coval and Stafford, 2007). Funds that view market timing as a priority in investment decisions could however value the liquidity provided by cash holding. We show in Table 3 that Flexible funds hold more cash than Rigid funds do, and sole-managed Flexible funds on average hold the highest fraction of their assets in cash. Similarly, sole-managed Flexible funds also exhibit the highest dispersion of cash holdings across funds, indicating that these funds could change cash holding levels based on market timing. Evidence on cash holdings thus is consistent with liquidity management by fund managers driven by cross-asset market timing trading strategies. Such liquidity management comes at a cost, and high level of cash holdings can provide a drag on fund performance because of the low returns generated in cash-equivalent securities (see Wermers, 2000).

5.3 Market Timing, Security Selection and Fund Flow

In this subsection, we examine the performance-flow relation in the context of market timing and security selection. The objective of this exercise is to further explore the management structure decisions of the fund families. We show in the previous subsection that market timing skills contribute positively to overall fund performance. One question that arises naturally is whether investors recognize market timing skills and whether the presence of market timing skills affects their decisions of investing in balanced funds. This question has important implications on the management structure decisions of the mutual fund companies. Depending on whether investors recognize market timing skills or whether they take market timing skills into consideration when making investment decisions, mutual fund families could structure fund management differently.

To analyze the flow-performance relation, we again use the market timing and security selection measures obtained from the performance attribution procedure, as well as the Treynor-Mazuy and Henriksson-Merton models. We use the three models to ensure robustness of our results. For each year, we sort funds within Rigid and Flexile categories into five quintiles based on market timing skills and security selection skills, respectively. We assign a high performance dummy to the funds in the top quintile (“Hi-MT” and “Hi-SS” for market timing and security selection, respectively) and assign a low performance dummy to the funds in the bottom quintile (“Lo-MT” and “Lo-SS” for market timing and security selection, respectively). We use the high and low performance dummies to capture the non-linear relation between fund performance and fund flows (see, Sirri and Tufano, 1998; Chevalier and Ellison, 1997). We compute fund flows first for each quarter and then obtain fund flow for each year by summing up the quarterly net fund flows.

The quarterly net flow is defined as the dollar change in total net assets (TNA) minus the appreciation in the fund assets. In the case of fund merger, we remove from fund asset value the increase in total assets due to merger. That is, the net cash flow to fund i during quarter q is:

$$\text{Flow}_{i,q} = \text{TNA}_{i,q} - \text{TNA}_{i,q-1} \times (1 + R_{i,q}) - \text{MGTNA}_{i,q}, \quad (9)$$

where $\text{TNA}_{i,q}$ is the total net assets of fund i at the end of quarter q , $R_{i,q}$ is its return during

quarter q , and $\text{MGTNA}_{i,q}$ is the increase in the total net assets of fund i due to mergers during quarter q . The annual fund flows are then scaled by fund asset value at the end of the previous year. To control for the impact of fund flow persistence and fund flow into the balanced fund sector, we include lagged fund flows and the average fund flow into the balanced fund sample in the regressions.

We estimate the following pooled regressions:

$$\text{Flow}_{i,t+1} = \alpha + \beta_{HS}\text{Hi-SS}_{i,t} + \beta_{LS}\text{Lo-SS}_{i,t} + \beta_{HM}\text{Hi-MT}_{i,t} + \beta_{LM}\text{Lo-MT}_{i,t} + \gamma\text{Controls} + \epsilon_{i,t}. \quad (10)$$

The control variables include lagged fund flows for fund i and average fund flow into the balanced fund sector during period $t+1$. Additionally, we include fund size (log value of total net assets), expense ratio, turnover, fund age, and the standard deviation of monthly fund returns over the past 36 months in the regression.

Table 10 reports the pooled regression results. We first run the regression using the full sample of funds, and then run the same regression separately for sole- and team-managed funds. We report the t -statistics calculated based on standard errors clustered at the fund level. Columns [1]-[3] employ market timing and security selection as measured from the performance attribution approach. In columns [4]-[6] and [7]-[9], we employ the measures obtained from the Treynor-Mazuy and Henriksson-Merton models, respectively.

Results from the three panels are consistent. In columns [1]-[3], results using all the funds suggest that investors value both market timing skills and security selection skills in their investment decisions. High performance in both market timing and security selection has a positive impact on fund flows, while poor performance negatively affects fund flows. Within the sole-managed fund sample, there seems to be a stronger flow-performance relation. In contrast, the team-managed funds show a much weaker impact of market timing on fund flows. Surprisingly, security selection skills also have negligible impact on fund flows for team-managed funds.

Results based on the Treynor-Mazuy and the Henriksson-Merton models are similar. In the full sample, both market timing skills and security selection skills affect investors' fund-investment decisions. The relation between performance, due to both market timing skills and

security selection skills, and fund flows holds strongly for sole-managed funds. There is also robust evidence that security selection affects fund flows for team-managed funds. However, the presence (or lack) of market timing skills has virtually no impact on the investors in the team-managed funds.

The flow-performance results show that investors do value market timing skills. The pattern is particularly strong for sole-managed funds. The lack of a positive relation between market timing and fund flow in the team-managed funds could be related to the following two considerations. First, as we show earlier, team-managed funds on average do not exhibit significant market timing skills. Second, investors may recognize that team-managed funds are less likely to engage in market timing activities, and thus put less emphasis on market timing skills when evaluating their performance.

Investors also react to security selection skills differently for sole- and team-managed funds. For sole-managed funds, investors react strongly to both negative and positive security selection performance, and the reaction to good performance is much stronger (as is consistent with the general mutual fund literature). In contrast, investor reaction to security selection performance of team-managed funds overall is weaker. Interestingly, investors react more strongly to bad security selection performance of team-managed funds than to good security selection performance. The weaker investor reaction to team-managed fund performance could be consistent with the argument in Massa, Reuter, and Zitzewitz (2010) that investors may be less likely to attribute the performance to the managers in team-managed funds than in sole-managed funds. Such an argument, however, is unlikely to completely explain the differences in market timing and fund flow relation between sole- and team-managed funds.

In this section, we conduct several tests in order to better understand the management's decisions on the structure of balanced funds, and how such decisions by the management affect fund investment decisions of investors. The results show that differences in market timing performance between sole- and team-managed balanced funds are not driven by possible differences in investment objectives of these funds. Analysis of market timing within equity portfolio suggests that both sole- and team-managed funds pursue market timing strategies. Evidence on fund flows also suggests that fund investors recognize and value market timing skills. We

also document that market timing investment strategies could adversely affect security selection decisions. The trade-off between the investment strategies, in combination with the trade-off between specialization benefits and coordination costs in teams, could help to explain the co-existence of team and sole-managed balanced funds.

6 Conclusion

We study how the allocation of control rights in mutual funds affects their investment strategies using a sample of balanced funds. We find that sole- and team-managed funds differ with respect to both investment strategies and investment performance. Sole-managed funds exhibit significant market timing ability while team-managed funds do not. On the other hand, team-managed funds show greater security-selection ability than sole-managed funds. We evaluate various alternative explanations for this evidence. Our analysis shows that coordination costs in teams, rather than managerial ability, explain the lack of market timing in team-managed funds.

The paper presents direct evidence on the coordination costs in teams or in decentralized organizations. The results provide empirical support for the argument that specialization benefits and coordination costs jointly determine the degree of specialization and the structure of teams (firms) (Becker and Murphy, 1992). In the context of balanced funds, the trade-off between specialization benefits and coordination costs in teams helps to explain the use of teams in mutual fund management, funds' investment strategies, and performance.

Our results have implications on the use of teams in the mutual fund industry. The evidence suggests that, there is little "joint" production in teams in the management of balanced funds because of the significant coordination costs. How important are coordination costs in fund management in the overall mutual fund industry? While it is conceivable that the coordination costs can be lower in mutual funds holding more homogeneous assets, that do not reallocate investments across asset classes (for example, equity or bond funds), such costs can still be substantial if the investment decisions are made jointly. Because investment research and decisions involve the processing of soft information (Stein, 2002), fund managers may have a harder time convincing others of their investment ideas and it becomes more difficult to

incorporate the soft information in joint investment decisions. Indeed, Chen *et. al.* (2004) find that sole-managed funds are significantly more likely than team-managed funds to invest in local stocks and to do better than team-managed funds at picking local stocks. Such findings are consistent with the notion that coordination costs can still play a significant role in the overall mutual fund industry.

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Table 1: Summary Statistics of the Balanced Fund Sample

This table reports summary statistics of the balanced funds in the sample. It lists the number of the funds in Panel A and the average fund size (Total Net Assets, TNA, in millions) in Panel B, for each year and for sole-managed (Sole) as well as team-managed (Team) funds. The summary statistics are first presented for the full balanced fund sample (All) and then separately for the Rigid and Flexible fund samples. Rigid funds maintain a balanced portfolio of both stocks and bonds within a pre-specified range. Flexible funds are generally free to assign up to 100% of their assets across various asset classes including domestic equities, fixed-income securities, and money market instruments.

Panel A: Number of Funds

Year	Full Sample			Rigid		Flexible	
	All	Sole	Team	Sole	Team	Sole	Team
1992	153	92	61	52	45	40	16
1993	201	121	80	69	61	52	19
1994	258	148	110	80	82	68	28
1995	268	151	117	81	88	70	29
1996	283	150	133	78	99	72	34
1997	314	147	167	79	123	68	44
1998	358	166	192	92	140	74	52
1999	360	148	212	82	151	66	61
2000	364	150	214	80	152	70	62
2001	355	139	216	73	155	66	61
2002	353	128	225	69	157	59	68
2003	355	112	243	61	161	51	82
2004	367	102	265	54	176	48	89
2005	351	93	258	50	171	43	87
2006	335	88	247	47	165	41	82
2007	316	83	233	45	158	38	75
2008	300	78	222	42	151	36	71
2009	274	68	206	36	138	32	68

Panel B: Average Fund Size (TNA)

Year	Full Sample			Rigid		Flexible	
	All	Sole	Team	Sole	Team	Sole	Team
1992	346	325	376	357	456	283	154
1993	433	430	438	444	492	411	268
1994	386	388	383	411	433	362	236
1995	485	402	593	333	705	481	253
1996	549	441	670	353	808	536	268
1997	633	474	772	372	892	594	435
1998	670	486	829	373	956	627	487
1999	704	524	829	442	954	626	520
2000	664	496	782	402	924	603	435
2001	680	516	786	470	931	568	417
2002	623	451	722	423	891	484	329
2003	813	599	911	574	1176	630	389
2004	951	723	1039	721	1352	725	419
2005	1109	851	1202	811	1557	896	504
2006	1319	1127	1387	1067	1740	1196	675
2007	1540	1331	1615	1255	1973	1422	862
2008	1128	965	1186	828	1429	1125	669
2009	1487	1267	1559	1132	1842	1420	985

Table 2: Summary Characteristics of Team- and Sole-managed Balanced Funds

This table reports characteristics of the sole- and team-managed balanced funds, and their differences. The average monthly returns and the standard deviations of monthly returns are first computed for each year for each fund, and then averaged across the fund categories (Sole- and Team-managed, Rigid and Flexible). The table reports the time series average of the mean value (in percentage) of monthly returns and standard deviations, expense ratio, and annual turnover ratio for sole- and team-managed funds for the overall sample (All Funds) as well as for Rigid and Flexible funds for the sample period of 1992-2009.

	All Funds	Sole-Managed	Team-Managed	Sole – Team	<i>t</i> -stat
Average Monthly Return					
All Funds	0.57	0.56	0.57	0.00	[-0.25]
Rigid Funds	0.56	0.57	0.56	0.01	[0.70]
Flexible Funds	0.57	0.56	0.59	-0.04	[-1.22]
FLX – RIG	0.01	-0.01	0.04		
<i>t</i> -stat	[0.27]	[0.24]	[0.21]		
Standard Deviation of Monthly Returns					
All Funds	2.63	2.74	2.53	0.21	[5.98]
Rigid Funds	2.52	2.63	2.47	0.16	[5.19]
Flexible Funds	2.79	2.87	2.68	0.19	[3.05]
FLX – RIG	0.53	-0.24	1.55		
<i>t</i> -stat	[7.99]	[5.51]	[3.90]		
Expense Ratio					
All Funds	1.20	1.27	1.15	0.12	[5.60]
Rigid Funds	1.09	1.17	1.04	0.13	[7.97]
Flexible Funds	1.30	1.37	1.26	0.11	[2.70]
FLX – RIG	0.21	0.20	0.22		
<i>t</i> -stat	[5.53]	[6.36]	[4.22]		
Turnover					
All Funds	0.92	0.87	0.94	-0.07	[-1.40]
Rigid Funds	0.80	0.72	0.84	-0.13	[-4.01]
Flexible Funds	1.04	1.03	1.04	-0.01	[-0.12]
FLX – RIG	0.24	0.31	0.19		
<i>t</i> -stat	[7.48]	[5.00]	[3.27]		

Table 3: Asset Allocation in Balanced Fund Portfolios

This table reports summary statistics of asset allocation among stocks, bonds, and cash by the balanced funds in the sample. We use the percentage of asset allocations at the end of each year to compute the mean and cross-sectional dispersion of asset allocation among the funds in the sample. We compute these statistics for sole- and team-managed funds, and for Rigid funds and Flexible funds within sole- and team-managed funds. Using these annual statistics, we report the time series average of the asset allocation percentages and the dispersions for sole- and team-managed funds, and their differences.

Panel A: Allocation in Stocks

	All	Sole-Managed	Team-Managed	Sole – Team	<i>t</i> -stat
Mean allocation					
All Funds	59.47	63.25	59.71	3.54	[5.78]
Rigid Funds	58.27	61.93	58.48	3.45	[8.17]
Flexible Funds	61.60	64.59	62.90	1.70	[1.43]
Dispersion across funds					
All Funds	18.41	19.39	16.35	3.05	[7.96]
Rigid Funds	14.51	14.28	13.03	1.25	[2.95]
Flexible Funds	23.41	23.72	22.14	1.58	[2.15]

Panel B: Allocation in Bonds

	All	Sole-Managed	Team-Managed	Sole – Team	<i>t</i> -stat
Mean allocation					
All Funds	29.36	25.23	30.94	-5.71	[-8.13]
Rigid Funds	33.09	29.96	33.31	-3.35	[-4.54]
Flexible Funds	22.87	19.92	24.81	-4.89	[-4.43]
Dispersion across funds					
All Funds	17.24	17.26	16.17	1.08	[3.90]
Rigid Funds	14.80	14.44	14.02	0.42	[0.83]
Flexible Funds	19.04	18.47	18.86	-0.39	[-0.70]

Panel C: Allocation in Cash

	All	Sole-Managed	Team-Managed	Sole – Team	<i>t</i> -stat
Mean allocation					
All Funds	11.14	11.48	9.33	2.15	[5.15]
Rigid Funds	8.61	8.07	8.20	-0.13	[-0.26]
Flexible Funds	15.51	15.45	12.28	3.16	[3.19]
Dispersion across funds					
All Funds	15.63	16.19	11.89	4.30	[4.00]
Rigid Funds	11.51	10.82	10.08	0.73	[0.77]
Flexible Funds	19.82	19.75	15.13	4.63	[2.74]

Table 4: Evidence of Market Timing in Balanced Funds

This table reports the market timing coefficients from the two-factor and five-factor Treynor-Mazuy and Henriksson-Merton market timing models. For each balanced fund in the sample, we estimate the market timing models over a three-year period based on monthly returns for each year. We then compute the average market timing coefficients for sole- and team-managed balanced funds, and for the two sub-samples of Rigid and Flexible funds for each year. The table reports the time series averages of the market timing coefficient estimates along with t -statistics (in brackets) that are based on autocorrelation-adjusted standard errors.

Panel A: Treynor-Mazuy Market Timing Model

Models	Full Sample		Sole-Managed		Team-Managed	
	Market Timing	t -stat	Market Timing	t -stat	Market Timing	t -stat
All Funds						
2-factor	0.0206	[0.84]	0.0770	[1.59]	0.0178	[0.68]
5-factor	0.0648	[2.62]	0.1483	[3.04]	0.0207	[0.81]
Rigid Funds						
2-factor	0.0115	[0.43]	0.0678	[1.13]	0.0101	[0.38]
5-factor	0.0452	[1.69]	0.1222	[2.04]	0.0296	[1.17]
Flexible Funds						
2-factor	0.0521	[1.07]	0.0873	[1.15]	0.0361	[0.59]
5-factor	0.0883	[1.91]	0.1775	[2.42]	-0.0003	[-0.01]

Panel B: Henriksson-Merton Market Timing Model

Models	Full Sample		Sole-Managed		Team-Managed	
	Market Timing	t -stat	Market Timing	t -stat	Market Timing	t -stat
All Funds						
2-factor	0.0030	[0.87]	0.0052	[0.76]	0.0061	[1.83]
5-factor	0.0064	[2.03]	0.0140	[2.12]	0.0022	[0.65]
Rigid Funds						
2-factor	0.0017	[0.47]	0.0049	[0.62]	0.0029	[0.81]
5-factor	0.0013	[0.37]	0.0094	[1.36]	-0.0002	[-0.07]
Flexible Funds						
2-factor	0.0101	[1.56]	0.0055	[0.49]	0.0136	[1.66]
5-factor	0.0158	[2.42]	0.0191	[1.82]	0.0080	[1.02]

Table 5: Management Structural Changes and Market Timing

This table presents evidence on changes of market timing and return performance for a sample of funds that changed their management structure from sole-management to team-management, or vice versa, over the sample period. The results are from the five-factor models without market timing (Five-factor) and the five-factor Treynor-Mazuy and Henriksson-Merton market timing models for the three-year periods before and after the year of management structure change. The table reports the value of estimated alphas and the market timing coefficients from the two market timing models and the differences (aft-bef) in fund performance as well as market timing around management structure change. *t*-statistics are reported in brackets.

Panel A: From Sole Management to Team Management

Model	Alpha			Market Timing			Market Timing		
	before	after	aft-bef	before	after	aft-bef	before	after	aft-bef
Five-factor (without market-timing)	-1.426 [-4.77]	-1.820 [-6.39]	-0.483 [-1.26]						
Treynor-Mazuy	-1.365 [-4.10]	-2.168 [-6.00]	-0.904 [-1.89]	0.43 [2.76]	0.05 [0.64]	-0.38 [-2.23]			
Henriksson-Merton	-1.113 [-2.75]	-2.500 [-5.52]	-1.441 [-2.37]				0.052 [2.62]	-0.007 [-0.53]	-0.057 [-2.49]

Panel B: From Team Management to Sole Management

Model	Alpha			Market Timing			Market Timing		
	before	after	aft-bef	before	after	aft-bef	before	after	aft-bef
Five-factor (without market-timing)	-2.665 [-2.66]	-0.131 [-0.25]	1.863 [2.29]						
Treynor-Mazuy	-5.782 [-3.43]	-1.078 [-1.89]	4.153 [2.66]	0.21 [1.38]	0.67 [3.08]	0.48 [2.26]			
Henriksson-Merton	-7.999 [-3.20]	-1.756 [-2.23]	5.909 [2.57]				0.053 [1.51]	0.112 [2.64]	0.069 [1.67]

Table 6: Market Timing, Security Selection and Fund Performance

This table reports the overall fund performance as well as its security selection (SS) and market timing (MT) components. We obtain the SS and MT performance from the five-factor Treynor-Mazuy and Henriksson-Merton market timing models, and compute the overall performance as the sum of SS and MT. The table reports the performance measures for sole- and team-managed funds within Rigid and Flexible funds, and the differences between sole- and team-managed funds. Panel A reports results based on gross returns (before expense) and Panel B reports results based on net returns (after expense). *t*-statistics (in brackets) are autocorrelation-adjusted.

Panel A: Gross Returns						
	Treynor-Mazuy Model			Henriksson-Merton Model		
	SS	MT	Overall	SS	MT	Overall
All Funds						
Rigid Funds	-0.056 [-0.59]	-0.086 [-1.94]	-0.142 [-2.08]	0.003 [0.02]	-0.088 [-1.13]	-0.085 [-1.18]
Flexible Funds	0.172 [0.98]	-0.107 [-1.10]	0.065 [0.57]	0.049 [0.21]	0.147 [0.83]	0.196 [1.59]
Sole-Managed Funds						
Rigid Funds	0.001 [0.00]	0.132 [1.36]	0.133 [0.78]	-0.003 [-0.01]	0.214 [1.16]	0.211 [1.15]
Flexible Funds	-0.186 [-0.65]	0.264 [1.99]	0.077 [0.98]	-0.214 [-1.29]	0.313 [1.88]	0.099 [0.51]
Team-Managed Funds						
Rigid Funds	-0.079 [-0.88]	-0.175 [-1.02]	-0.254 [-3.84]	0.005 [0.04]	-0.211 [-2.68]	-0.206 [-2.98]
Flexible Funds	0.503 [2.46]	-0.356 [-2.78]	0.146 [0.96]	0.292 [1.13]	-0.007 [-0.04]	0.285 [1.85]
Sole – Team						
Rigid Funds	0.080 [0.88]	0.307 [1.92]	0.387 [2.02]	-0.008 [-0.03]	0.425 [1.85]	0.416 [1.80]
Flexible Funds	-0.689 [-2.64]	0.620 [3.89]	-0.069 [0.04]	-0.505 [-1.71]	0.320 [1.75]	-0.186 [-1.12]
Panel B: Net Returns						
	Treynor-Mazuy			Henriksson-Merton		
	SS	MT	Overall	SS	MT	Overall
All Funds						
Rigid Funds	-1.138 [-11.68]	-0.081 [-1.80]	-1.220 [-17.37]	-1.081 [-9.04]	-0.077 [-0.98]	-1.158 [-15.62]
Flexible Funds	-1.072 [-5.98]	-0.129 [-1.31]	-1.201 [-10.22]	-1.175 [-4.94]	0.100 [0.55]	-1.075 [-8.56]
Sole-Managed Funds						
Rigid Funds	-1.253 [-5.11]	0.156 [1.57]	-1.096 [-6.35]	-1.282 [-4.32]	0.277 [1.45]	-1.005 [-5.41]
Flexible Funds	-1.527 [-5.11]	0.238 [0.92]	-1.289 [-5.16]	-1.542 [-3.78]	0.278 [0.87]	-1.264 [-6.26]
Team-Managed Funds						
Rigid Funds	-1.093 [-11.48]	-0.176 [-3.59]	-1.268 [-18.11]	-1.001 [-8.45]	-0.218 [-2.74]	-1.219 [-16.76]
Flexible Funds	-0.654 [-3.19]	-0.375 [-2.91]	-1.029 [-6.73]	-0.837 [-3.22]	-0.065 [-0.36]	-0.902 [-5.87]
Sole – Team						
Rigid Funds	-0.160 [-1.87]	0.332 [2.54]	0.172 [1.57]	-0.281 [-1.88]	0.495 [2.44]	0.214 [1.16]
Flexible Funds	-0.873 [-1.89]	0.612 [3.34]	-0.260 [-1.56]	-0.705 [-2.81]	0.343 [2.10]	-0.362 [-1.24]

Table 7: Market Timing within Equity Portfolios

This table reports the estimated market timing coefficients from the Treynor-Mazuy and Henriksson-Merton market timing models based on fund stock holdings. The dependent variables in both models are the fund betas estimated from the fund equity holdings, and the independent variables are the square of market excess returns and the absolute value of market excess returns from the Treynor-Mazuy and the Henriksson-Merton market timing models, respectively. The table reports the average coefficient estimates and their t -statistics (in brackets) for the team- and sole-managed balanced funds. Panel A reports results using Rigid funds and Panel B reports results based on Flexible funds.

Panel A: Rigid Funds

Treynor-Mazuy Model			Henriksson-Merton Model	
Sole-Managed	Intercept	1.010 [44.32]	Intercept	0.978 [27.45]
	Market Timing ($\gamma_{i,TM}$)	0.366 [2.21]	Market Timing ($\gamma_{i,HM}$)	0.058 [1.89]
Team-Managed	Intercept	0.996 [54.67]	Intercept	0.972 [33.33]
	Market Timing ($\gamma_{i,TM}$)	0.289 [2.45]	Market Timing ($\gamma_{i,HM}$)	0.045 [1.96]

Panel B: Flexible Funds

Treynor-Mazuy Model			Henriksson-Merton Model	
Sole-Managed	Intercept	1.025 [53.23]	Intercept	1.000 [31.93]
	Market Timing ($\gamma_{i,TM}$)	0.276 [1.87]	Market Timing ($\gamma_{i,HM}$)	0.046 [1.78]
Team-Managed	Intercept	1.016 [41.02]	Intercept	0.986 [25.60]
	Market Timing ($\gamma_{i,TM}$)	0.336 [2.08]	Market Timing ($\gamma_{i,HM}$)	0.054 [1.88]

Table 9: Stock Holdings and Portfolio Liquidity

Panels A and B of this table report the number of stocks in the equity portfolio and the size of the equity portfolio, respectively, for sole- and team-managed balanced funds. The number and value of stocks are computed each quarter for each fund. We obtain the mean and median values for sole- and team-managed Rigid and Flexible funds for each quarter. The table reports the time series average of the mean and median values for the two types of funds and their differences. Panel C reports the value- and equal- weighted Amihud (2002) (il)liquidity measures of stock portfolios for sole and team managed funds and their differences. We compute the (il)liquidity measures for each fund in each quarter and then obtain the average across the funds in a quarter. The table reports a logarithmic transformation of the time series average of these quarterly mean values. *t*-statistics are reported in brackets.

Panel A: Number of Stocks in the Equity Portfolio						
	Mean			Median		
	Sole	Team	Sole–Team	Sole	Team	Sole–Team
Rigid Funds	63.14	157.83	-94.69 [-16.96]	48.30	70.35	-22.06 [-12.80]
Flexible Funds	137.86	211.01	-73.15 [-8.72]	68.49	104.40	-35.90 [-6.29]

Panel B: Value of Stock Holdings						
	Mean			Median		
	Sole	Team	Sole–Team	Sole	Team	Sole–Team
Rigid Funds	257.05	661.47	-404.42 [-7.66]	72.21	78.53	-6.33 [-1.32]
Flexible Funds	439.39	282.60	156.79 [6.45]	89.61	84.68	4.93 [0.40]

Panel C: Liquidity of Stock Holdings						
	Mean			Median		
	Sole	Team	Sole–Team	Sole	Team	Sole–Team
Rigid Funds	0.021	0.020	0.001 [3.01]	0.025	0.027	-0.003 [-6.30]
Flexible Funds	0.024	0.024	0.000 [-0.11]	0.033	0.034	-0.001 [-1.30]

Table 10: Security Selection, Market Timing and Fund Flow

This table reports pooled OLS regression estimates of the sensitivity of fund flows to performance. For each year, we sort balanced funds within the Rigid and Flexible categories into five quintiles based on market timing skills and security selection skills, respectively. The market timing and security selection measures are obtained from the performance attribution procedure, the Treynor-Mazuy, and Henriksson-Merton market timing models. We assign a high performance dummy to the funds in the top quintile for market timing and security selection respectively (*Hi-MT* and *Hi-SS*), and assign a low performance dummy to the funds ranked in the bottom quintile (*Lo-MT* and *Lo-SS*). Fund flows are annual money flow into the fund scaled by the fund's asset value at the beginning of the year. The control variables include lagged fund flow in the previous year (*Lag Flow*), average fund flow into the balanced fund sector (*Style Flow*), fund size (*Log(TNA)*, logarithm of total fund assets), expense ratio (*Expense*), turnover (*Turnover*), fund age (*Log(Age)*, logarithm of fund age), and the standard deviation of monthly fund returns over the past three years (*Std Dev*). [*t*-statistics] are based on standard errors clustered at fund level.

	Performance Attribution				Treynor-Mazuy Model				Henriksson-Merton Model			
	All Funds	Sole	Team		All Funds	Sole	Team		All Funds	Sole	Team	
	[1]	[2]	[3]		[4]	[5]	[6]		[7]	[8]	[9]	
Hi-SS	0.054 [1.89]	0.105 [2.71]	0.032 [0.77]		0.141 [3.43]	0.210 [3.96]	0.093 [2.42]		0.100 [2.96]	0.220 [3.42]	0.039 [1.25]	
Lo-SS	-0.065 [-2.82]	-0.131 [-2.87]	-0.028 [-1.23]		-0.092 [-5.07]	-0.087 [-3.02]	-0.089 [-5.61]		-0.034 [-1.49]	-0.020 [-0.47]	-0.034 [-2.09]	
Hi-MT	0.060 [3.18]	0.132 [3.24]	0.013 [0.56]		0.054 [2.04]	0.071 [2.01]	0.043 [1.33]		0.034 [2.50]	0.100 [2.90]	-0.009 [-0.71]	
Lo-MT	-0.032 [-1.26]	-0.060 [-1.31]	-0.018 [-0.63]		0.000 [0.02]	-0.007 [-0.12]	-0.005 [-0.19]		-0.013 [-0.50]	-0.039 [-0.66]	-0.001 [-0.04]	
Log(TNA)	-0.006 [-1.88]	-0.007 [-1.31]	-0.007 [-2.40]		-0.005 [-1.50]	-0.006 [-1.08]	-0.006 [-2.23]		-0.006 [-1.65]	-0.005 [-0.96]	-0.007 [-2.29]	
Expense	-0.785 [-0.39]	-2.863 [-1.52]	0.132 [0.06]		-1.982 [-1.26]	-3.892 [-2.08]	-0.791 [-0.49]		-0.648 [-0.30]	-2.864 [-1.25]	0.555 [0.25]	
Turnover	-0.007 [-0.71]	-0.005 [-0.79]	-0.010 [-0.52]		-0.005 [-0.59]	-0.004 [-0.63]	-0.007 [-0.40]		-0.007 [-0.70]	-0.009 [-1.64]	-0.010 [-0.52]	
Log(Age)	0.009 [0.88]	-0.006 [-0.52]	0.014 [1.49]		0.006 [0.60]	-0.005 [-0.41]	0.012 [1.28]		0.006 [0.58]	-0.011 [-0.91]	0.014 [1.49]	
Std. Dev.	-0.518 [-0.47]	1.095 [0.58]	-1.382 [-1.82]		-0.348 [-0.36]	1.057 [0.61]	-1.159 [-1.60]		-0.830 [-0.78]	0.140 [0.08]	-1.467 [-1.83]	
Lag Flow	0.333 [4.14]	0.150 [2.04]	0.502 [7.47]		0.313 [4.05]	0.146 [1.89]	0.477 [7.39]		0.319 [4.02]	0.122 [1.69]	0.494 [7.44]	
Style Flow	0.879 [9.20]	0.934 [6.93]	0.832 [5.97]		0.894 [11.31]	0.915 [6.22]	0.837 [7.42]		0.900 [9.87]	0.993 [5.91]	0.842 [5.96]	
Intercept	0.032 [0.72]	0.030 [0.46]	0.046 [1.05]		0.024 [0.52]	0.012 [0.16]	0.035 [1.05]		0.028 [0.62]	0.019 [0.30]	0.046 [1.02]	
Observations	2,099	741	1,358		2,099	741	1,358		2,099	741	1,358	
R ²	13.9%	8.2%	24.8%		17.5%	10.2%	26.5%		14.5%	9.1%	24.4%	